

# Task 4: SQL for Data Analysis

## Dataset: Bike rental Store Sharing

| ride_date           | ride_count | temp_1 | temp_2 | humidity | wind | weather | holiday_flag | weekend_flag | season_number |
|---------------------|------------|--------|--------|----------|------|---------|--------------|--------------|---------------|
| 2015-01-04 00:00:00 | 182        | 3      | 2      | 93       | 6    | 3       | 0            | 1            | 3             |
| 2015-01-04 01:00:00 | 138        | 3      | 2.5    | 93       | 5    | 1       | 0            | 1            | 3             |
| 2015-01-04 02:00:00 | 134        | 2.5    | 2.5    | 96.5     | 0    | 1       | 0            | 1            | 3             |
| 2015-01-04 03:00:00 | 72         | 2      | 2      | 100      | 0    | 1       | 0            | 1            | 3             |
| 2015-01-04 04:00:00 | 47         | 2      | 0      | 93       | 6.5  | 1       | 0            | 1            | 3             |
| 2015-01-04 05:00:00 | 46         | 2      | 2      | 93       | 4    | 1       | 0            | 1            | 3             |
| 2015-01-04 06:00:00 | 51         | 1      | -1     | 100      | 7    | 4       | 0            | 1            | 3             |
| 2015-01-04 07:00:00 | 75         | 1      | -1     | 100      | 7    | 4       | 0            | 1            | 3             |

| ride_date           | wind | ride_count |
|---------------------|------|------------|
| 2015-11-17 21:00:00 | 56.5 | 644        |
| 2016-02-08 15:00:00 | 56   | 556        |
| 2016-03-28 08:00:00 | 54   | 65         |
| 2016-03-28 06:00:00 | 52   | 47         |
| 2016-02-08 16:00:00 | 51   | 964        |
| 2015-11-17 20:00:00 | 51   | 934        |
| 2016-03-28 09:00:00 | 50   | 169        |
| 2016-02-08 12:00:00 | 49.5 | 612        |

| season_number | total_rides |
|---------------|-------------|
| 0             | 4850236     |
| 1             | 6424609     |
| 2             | 5073040     |
| 3             | 3558087     |

| max_temp | min_temp | avg_temp          |
|----------|----------|-------------------|
| 34       | -1.5     | 12.46809080814376 |

| weekend_flag | avg_rides |
|--------------|-----------|
| 1            | 977.4157  |
| 0            | 1209.2748 |

| weather | total_rides |
|---------|-------------|
| 1       | 7146847     |
| 2       | 6035580     |
| 3       | 4243887     |
| 7       | 1526461     |
| 4       | 929978      |
| 26      | 15051       |
| 10      | 8168        |

| ride_date           | ride_count |
|---------------------|------------|
| 2015-01-05 07:00:00 | 1597       |
| 2015-01-05 08:00:00 | 3161       |
| 2015-01-05 09:00:00 | 1342       |
| 2015-01-05 16:00:00 | 1192       |
| 2015-01-05 17:00:00 | 2761       |
| 2015-01-05 18:00:00 | 2389       |
| 2015-01-05 19:00:00 | 1274       |
| 2015-01-06 07:00:00 | 1715       |

|   | id | select_type | table          | partitions | type  | possible_keys     | key               | key_len | ref  | rows  | filtered | Extra |
|---|----|-------------|----------------|------------|-------|-------------------|-------------------|---------|------|-------|----------|-------|
| ► | 1  | SIMPLE      | bike_rides_new | NULL       | index | idx_season_number | idx_season_number | 5       | NULL | 17031 | 100.00   | NULL  |

| month   | total_rides | avg_temp           | avg_humidity       |
|---------|-------------|--------------------|--------------------|
| 2015-01 | 546639      | 6.11177347242921   | 76.0134128166915   |
| 2015-02 | 543594      | 5.5461997019374065 | 77.42175856929956  |
| 2015-03 | 695934      | 8.245602165087957  | 70.13836265223274  |
| 2015-04 | 831582      | 10.949930458970792 | 67.41029207232268  |
| 2015-05 | 895413      | 13.563844086021506 | 64.43413978494624  |
| 2015-06 | 1033252     | 16.947698744769873 | 61.88493723849373  |
| 2015-07 | 1120687     | 19.063513513513513 | 63.016891891891895 |
| 2015-08 | 1033951     | 18.589381720430108 | 69.77620967741936  |

| #    | Time     | Action                                                                                | Message                                                        | Duration / Fetch      |
|------|----------|---------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------|
| ✓ 1  | 18:12:18 | USE bike_rental                                                                       | 0 row(s) affected                                              | 0.015 sec             |
| ✓ 2  | 18:12:37 | DROP TABLE IF EXISTS bike_rides_new                                                   | 0 row(s) affected                                              | 0.062 sec             |
| ✓ 3  | 18:12:46 | CREATE TABLE bike_rides_new AS SELECT timestamp AS ride_date, count AS ride_c...      | 17414 row(s) affected Records: 17414 Duplicates: 0 Warnings: 0 | 1.265 sec             |
| ✓ 4  | 18:12:59 | SELECT * FROM bike_rides_new LIMIT 0, 1000                                            | 1000 row(s) returned                                           | 0.016 sec / 0.000 sec |
| ✓ 5  | 18:14:32 | SELECT ride_date, ride_count FROM bike_rides_new WHERE ride_count > 1000 ORDER B...   | 1000 row(s) returned                                           | 0.015 sec / 0.016 sec |
| ✓ 6  | 18:14:32 | SELECT ride_date, wind, ride_count FROM bike_rides_new ORDER BY wind DESC LIMIT 10    | 10 row(s) returned                                             | 0.015 sec / 0.000 sec |
| ✓ 7  | 18:14:49 | SELECT DISTINCT season_number FROM bike_rides_new LIMIT 0, 1000                       | 4 row(s) returned                                              | 0.031 sec / 0.000 sec |
| ✓ 8  | 18:14:49 | SELECT season_number, SUM(ride_count) AS total_rides FROM bike_rides_new GROUP BY ... | 4 row(s) returned                                              | 0.047 sec / 0.000 sec |
| ✓ 9  | 18:15:04 | SELECT weekend_flag, AVG(ride_count) AS avg_rides FROM bike_rides_new GROUP BY w...   | 2 row(s) returned                                              | 0.032 sec / 0.000 sec |
| ✓ 10 | 18:15:12 | SELECT MAX(temp_1) AS max_temp, MIN(temp_1) AS min_temp, AVG(temp_1) AS avg_temp...   | 1 row(s) returned                                              | 0.016 sec / 0.000 sec |
| ✓ 11 | 18:15:20 | SELECT weather, SUM(ride_count) AS total_rides FROM bike_rides_new GROUP BY weathe... | 7 row(s) returned                                              | 0.047 sec / 0.000 sec |
| ✓ 12 | 18:15:41 | SELECT ride_date, ride_count FROM bike_rides_new WHERE ride_count > ( SELECT AV...    | 1000 row(s) returned                                           | 0.032 sec / 0.000 sec |
| ✓ 13 | 18:15:55 | DROP VIEW IF EXISTS monthly_summary                                                   | 0 row(s) affected                                              | 0.016 sec             |
| ✓ 14 | 18:16:08 | CREATE VIEW monthly_summary AS SELECT DATE_FORMAT(ride_date, '%Y-%m') AS mont...      | 0 row(s) affected                                              | 0.016 sec             |
| ✓ 15 | 18:16:18 | SELECT * FROM monthly_summary ORDER BY month LIMIT 0, 1000                            | 25 row(s) returned                                             | 0.078 sec / 0.000 sec |
| ✓ 16 | 18:17:43 | CREATE INDEX idx_ride_date ON bike_rides_new(ride_date)                               | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0         | 0.203 sec             |
| ✓ 17 | 18:17:43 | CREATE INDEX idx_season_number ON bike_rides_new(season_number)                       | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0         | 0.188 sec             |
| ✓ 18 | 18:18:04 | SELECT season_number, SUM(ride_count) AS total_rides FROM bike_rides_new GROUP BY ... | 4 row(s) returned                                              | 0.062 sec / 0.000 sec |
| ✓ 19 | 18:18:42 | EXPLAIN SELECT season_number, SUM(ride_count) AS total_rides FROM bike_rides_new G... | 1 row(s) returned                                              | 0.015 sec / 0.000 sec |

*Thank  
you!*